Research and development of algorithms for detecting distribution shifts in data based on data meta-features

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Abstract

Tabular out-of-distribution (OOD) detection asks whether a new row, or a batch of rows, follows the same generating distribution as a labelled in-distribution (ID) sample. This work studies the problem at two resolutions. A sample-wise procedure trains an autoencoder on ID rows and compares reconstruction-error distributions with a Kolmogorov–Smirnov test. Across six source–target pairs the empirical false-positive rate stays near the nominal level $\alpha = 0.05$. A point-wise extension summarizes reconstruction residuals with geometric descriptors and trains a supervised classifier. The partition method replaces one global reconstruction context with an ensemble of local reference cells. Quantile bins, K-means clusters, and randomized tree leaves define the cells. Each row is encoded by 36 meta-features that describe cell support, marginal shape, feature dependence, information content, and the position of the row inside the cell. Five-fold experiments on six pairs show that this encoding lets logistic regression reach ROC–AUC 0.998 on Taxi and 1.000 on ACS Accidents, values that a linear model on standardized raw attributes does not attain. The same encoding reduces the ROC–AUC range across logistic regression, a multilayer perceptron, and a random forest from 0.506 to 0.002 on Taxi and from 0.477 to 0.001 on ACS Accidents. Group ablations attribute the Taxi, Electricity, California, and ACS Accidents signals primarily to point-to-cell geometry, while Income and MVx6 load on marginal and dependence descriptors. The representation therefore supplies a linear, interpretable coordinate system for a known source–target shift. Code and data splits are available at `https://github.com/BodBodBod/ood_detection_itmo`.

Keywords: out-of-distribution detection, tabular data, meta-features, partition ensemble, supervised detection, distribution shift.

1 Introduction

Predictive systems are commonly trained under the assumption that future observations follow the training distribution. When this assumption fails, a model can keep returning confident outputs while the underlying assignment of labels has changed. Out-of-distribution detection provides a separate decision rule for rows or batches that are inconsistent with the available ID sample [35]. The decision is consequential in tabular applications such as finance, healthcare, industrial monitoring, and public-sector analytics, where a row mixes numerical and categorical attributes and carries little semantics outside its statistical context.

The same table can exhibit several kinds of change at once: a shift in one margin, a change in dependence between attributes, a local density change, or a displacement of a small subset of rows. A single global distance compresses these effects into one number. The resolution of the decision also matters. A sample-wise test can report that a batch has changed and still leave the responsible rows unidentified. A point-wise rule assigns a score to each row and therefore needs a description of that row relative to a reference distribution.

This study follows a progression from sample-wise testing to point-wise classification. The first stage trains an autoencoder on ID rows and compares distributions of reconstruction error with the two-sample Kolmogorov–Smirnov (KS) test. The KS statistic is the supremum gap between two empirical distribution functions. The stage asks whether reconstruction behaviour detects a contaminated batch at a controlled false-positive rate. The second stage summarizes each residual vector with norms, exceedance counts, and a Mahalanobis distance, then trains a binary classifier. The third stage builds an ensemble of local ID cells and represents every row by 36 named meta-features of those cells.

The supervised point-wise setting used for the partition method is specific. Training sees labelled rows from one source table (ID) and from one paired target table (OOD). Evaluation uses held-out rows from the same pair. The task is therefore discrimination of an observed source–target shift, sometimes called near-OOD detection when the two tables share the same attribute schema. The protocol does not test detection of an OOD type that is absent from training.

Three questions organize the empirical work.

1. Does an autoencoder combined with a KS test keep the prescribed type-I error and detect fully OOD batches?
2. Do geometric descriptors of reconstruction residuals improve point-wise ranking over the raw residual vector on matched splits?
3. Does a partition-based meta-feature encoding turn the source–target decision into a problem that a linear classifier can solve, and do the meta-feature groups identify which local statistics carry the shift?

The scientific novelty of the work lies in the third question. Existing supervised tabular detectors typically feed a classifier with raw attributes, a latent neural vector, reconstruction residuals, or a small set of complete anomaly scores [36, 21]. The present encoding keeps cell support, marginal shape, dependence, information content, and point-to-cell geometry as separate coordinates across many partitions. The coordinates have a fixed meaning from one table to another. The experiments measure two consequences of this choice: the quality of logistic regression on the encoding, and the spread of ROC–AUC across three classifier families.

The empirical contributions are as follows.

1. A sample-wise AE–KS procedure is validated by Monte Carlo simulation on six source–target pairs. Empirical false-positive rates lie between 0.044 and 0.050.
2. A supervised point-wise baseline on residual descriptors improves the best ROC–AUC relative to the raw residual vector on Taxi, Electricity, and Income.
3. A 36-dimensional partition encoding, evaluated with five-fold cross-validation on six pairs, yields logistic-regression ROC–AUC 0.998 on Taxi and 1.000 on ACS Accidents. On the same pairs the ROC–AUC range across logistic regression, a multilayer perceptron (MLP), and a random forest shrinks to 0.002 and 0.001. Group ablations and coefficient magnitudes associate Taxi, Electricity, California, and ACS Accidents with point-to-cell geometry, and associate Income and MVx6 with marginal and dependence descriptors.

2 Related works

2.1 Unsupervised OOD detection

Unsupervised OOD detection estimates normality from ID observations without labelled OOD examples [35]. Detectors differ in the statistical object used to define that normality.

Distance and neighborhood scores. Mahalanobis scoring fits class-conditional Gaussian statistics and measures covariance-normalized distance to a class centroid [13]. Relative Mahalanobis distance removes a class-independent background component for near-OOD observations [26]. Nearest-neighbor scoring uses distance to the k-th training neighbor in a learned representation [33, 25].

Minimum covariance determinant estimation supplies a robust Gaussian reference for the same family of distances [28]. Local Outlier Factor (LOF) compares the density around a point with densities around its neighbors [2]. Isolation Forest measures how readily random partitions isolate a point [16]. Histogram-based scoring aggregates per-feature tail ranks [7].

Distribution and density scores. Energy-based scoring derives a scalar from model logits [17, 4]. ECOD aggregates empirical per-feature tail probabilities for tabular rows [15]. COPOD models multivariate tails through an empirical copula [14]. Deep SVDD learns a hypersphere that encloses ID representations [29]. Autoencoder reconstruction error is a long-standing one-class score for multivariate tables [31, 37]. The sample-wise method in this paper uses that reconstruction residual as the input to a two-sample test.

Self-supervised representations. GOAD discriminates random affine transformations of general data [1]. NeuTraL AD learns transformations through a contrastive objective [24]. Internal Contrastive Learning agrees a tabular row with masked subsets of its features [32]. Deep noise evaluation recovers the magnitude of perturbations applied to normal tabular rows [3]. These methods learn structure from ID data alone.

Batch tests and concept drift. A change in a stream of tables is the subject of concept-drift detection [5, 19]. Two-sample tests such as maximum mean discrepancy [8] and the classifier two-sample test [18] decide whether two batches share a distribution. The AE–KS stage occupies this batch-level slot and uses reconstruction error as the tested coordinate.

Meta-features for distribution characterization. Meta-features summarize statistical properties of a collection rather than the raw coordinates of one row [27, 12]. Moments, information measures, and dependence statistics encode changes in a generating distribution. The present method computes such descriptors on many local ID cells and concatenates them with the position of the query row inside each cell.

2.2 Supervised OOD detection

Supervised and weakly supervised detectors use labelled anomalies or auxiliary OOD rows to learn an anomaly score [21]. Outlier Exposure trains with auxiliary outliers drawn from a related domain [10]. BORE treats unsupervised detector outputs as features for an imbalance-aware ensemble [20]. XGBOD selects informative detector scores, concatenates them with the original attributes, and fits XGBoost [36]. Deviation Networks, Deep SAD, FeaWAD, PReNet, and RoSAS learn from a small labelled anomaly subset [23, 30, 38, 22, 34]. ADBench documents large variation of these methods across tabular datasets and anomaly types [9]. Benchmarks such as TableShift organize tabular distribution shifts as explicit source–target pairs [6]. The pairs used here come from the ITMO NSS Lab tabular OOD collection [11].

The cited supervised methods learn from raw attributes, latent neural vectors, reconstruction residuals, or complete scores of base detectors. Local distributional structure of a heterogeneous table is typically collapsed into one score before the supervised stage. The gap addressed by the partition encoding is the retention of cell-level support, shape, dependence, information, and geometry as separate supervised inputs.

3 Problem setting

Let $\mathcal{D}_{\mathrm{ID}} = \{x_i\}_{i=1}^{n}$ be an ID reference sample with $x_i \in \mathbb{R}^d$ and let $\mathcal{D}_{\mathrm{OOD}}$ be a labelled sample from a paired target table that shares the same d attributes. A sample-wise detector receives a batch B of m rows and returns a binary decision "B differs from $\mathcal{D}_{\mathrm{ID}}$". A point-wise detector receives a single row x and returns a score $s(x) \in [0, 1]$, with larger values assigned to the OOD class.

For the partition method the training fold contains $\mathcal{D}_{\mathrm{ID}}^{\mathrm{train}}$ and $\mathcal{D}_{\mathrm{OOD}}^{\mathrm{train}}$. Partition functions and cell statistics are fitted on the ID training rows. The classifier is fitted on labelled ID and OOD training rows. Held-out rows from both classes form the test fold. Success is measured by ROC–AUC, which integrates true-positive rate over false-positive rate, by PR–AUC (average precision), and by F1 at the default probability threshold 0.5. ROC–AUC is the primary ranking metric. PR–AUC and F1 react more strongly to class imbalance.

Several pairs used in the study are already linearly or tree-separable on raw attributes. A random forest on standardized raw features reaches ROC–AUC 0.9999 on Taxi and 1.000 on ACS Accidents. Those pairs document the behaviour of a linear model after the encoding. Income and MVx6 leave more room between a constant classifier and a perfect ranking, and they are the pairs on which a change in representation is easier to interpret as a change in difficulty.

4 Partition meta-feature method

4.1 Meta-features of a reference cell

Each input coordinate is standardized with the ID training mean and standard deviation,

$$\widetilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}. \tag{1}$$

All subsequent definitions use this standardized space.

A partitioning function $P_m : \mathbb{R}^d \to \mathcal{A}_m$ maps a row to a cell identifier. The identifier is a quantile-bin index, a cluster index, or a decision-tree leaf. The ensemble $\mathcal{P} = \{P_m\}_{m=1}^{M}$ contains one quantile partition with ten bins for every feature, $\max(1, \lfloor d/2 \rfloor)$ K-means partitions with ten clusters on random feature subsets, and five randomized decision trees of depth three. The trees are fitted to independent uniform binary labels. Their role is to produce reproducible axis-aligned splits of the ID cloud. The resulting number of functions is $M = d + \max(1, \lfloor d/2 \rfloor) + 5$.

After P_m has been fitted on ID training rows, the reference cell of a query x is

$$C_m(x) = \left\{ z \in \mathcal{D}_{\mathrm{ID}}^{\mathrm{train}} \mid P_m(z) = P_m(x) \right\}. \tag{2}$$

$C_m(x)$ contains only ID training rows that received the same identifier as x. Cell means, covariances, entropies, and neighbor graphs are estimated on this set. A test row never enters the cell statistics.

The local descriptor vector is

$$\phi_m(x) = \left[h_1(x, C_m(x)), \ldots, h_q(x, C_m(x))\right]^{\top}, \qquad q = 36, \tag{3}$$

where each h_k is one entry of Table 1. Three coordinates illustrate the construction:

$$\log(1 + |C_m(x)|), \qquad \frac{|C_m(x)|}{n}, \qquad \|\widetilde{x} - \mu_m(x)\|_2 \,. \tag{4}$$

They record cell size, relative density, and Euclidean distance to the cell mean. The remaining coordinates apply the same pattern to robust moments, information quantities, dependence measures, and nearest-neighbor geometry. The quantity named `mahalanobis` uses the diagonal of the cell covariance. The quantity named `local_outlier_factor_score` is the ratio of the query k-nearest-neighbour distance to the mean k-nearest-neighbour distance inside the cell.

The partition representation concatenates the local vectors,

$$\Phi(x) = \left[\phi_1(x)^{\top}, \ldots, \phi_M(x)^{\top}\right]^{\top}. \tag{5}$$

Its length is Mq before optional raw features are appended. Figure 1 summarizes the data flow.

4.2 Supervised detector

Algorithm 1 states the fit and transform stages. For an ID or OOD query the fitted functions P_m return cell identifiers, the stored ID cell $C_m(x)$ supplies its statistics, and the blocks $\phi_m(x)$ are concatenated. An OOD row can land in an ID cell. Unusual distance, density, or neighbor values then appear as coordinates of $\Phi(x)$.

The vector can be used alone or concatenated with the standardized raw row,

$$\Psi(x) = \left[\Phi(x)^{\top}, \widetilde{x}^{\top}\right]^{\top}. \tag{6}$$

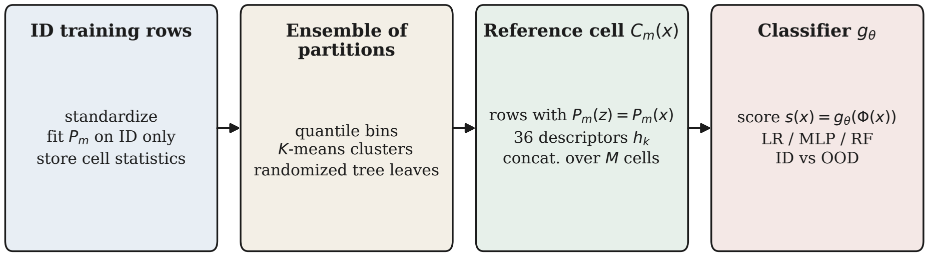

Figure 1: Construction of the partition representation. Each partition assigns the query to an ID training cell. Meta-features describe that cell and the position of the query inside it. The classifier is trained on the concatenated blocks.

Table 1: Groups of the 36 meta-features. The full list with formulae is given in Appendix A.

Group	Count	Information encoded
Point-based	8	Position of x in the cell: distances, diagonal-Gaussian log-likelihood, neighbor ratio, k-th neighbor distance, out-of-range count
Marginal statistical	7	Norms of cell mean, standard deviation, skewness, kurtosis, MAD, IQR, and trimmed mean
Marginal informational	5	Cell entropy, mean and spread of marginal entropies, mean marginal KL divergence to the ID reference, quantile surprisal
Interaction statistical	6	Covariance trace, log-determinant and condition number, mean absolute Pearson and Spearman correlations, diagonal Mahalanobis distance
Interaction informational	5	Mean and maximum pairwise mutual information, total correlation, mean pairwise joint entropy, mean Jensen–Shannon divergence
Base	5	Log cell size, density, coordinates beyond two standard deviations, cross-partition agreement, tree-leaf depth

A probabilistic classifier g_θ is trained with $y = 0$ for ID and $y = 1$ for OOD. The score is $s(x) = g_\theta(\Psi(x))$. Representation scaling is estimated on training ID vectors. The minority class is oversampled inside the training fold. Logistic regression, a random forest, and a two-hidden-layer MLP are trained on every representation. Agreement among these three models is part of the measured outcome.

5 Experimental setup

Data. The study uses six paired tables from the ITMO NSS Lab collection [11]. A source file is treated as ID and the matching target file as OOD. Table 2 reports sizes after the accompanying loaders. Taxi is loaded with six numeric attributes: the loader drops the unused index column of the released CSV.

Autoencoder experiments. An autoencoder is trained on ID rows after ID-only standardization. The encoder uses hidden widths 64 and 32 and a three-dimensional latent layer. The decoder mirrors this stack. Training minimizes mean squared reconstruction error with Adam at learning rate 10^{-3}. ID rows are split $70\%/15\%/15\%$ into train, validation, and test. Early stopping uses patience 30. California and ACS Accidents use a 50% ID training split.

The sample-wise experiment compares scalar reconstruction-error distributions. At $\alpha = 0.05$, each of $10{,}000$ Monte Carlo repetitions draws two ID bootstrap samples and one OOD sample of $1{,}000$ rows. A false positive is a KS rejection between the two ID error samples. A true positive is a

Algorithm 1 Fit and transform of the partition representation

Require: ID training matrix X_{ID}, query matrix X

Standardize columns of X_{ID} and store (μ, σ)
Fit $\{P_m\}_{m=1}^{M}$ on the standardized ID matrix
for $m = 1$ to M **do**
  Assign ID rows to cells and store statistics of each cell
end for
for each query x in X **do**
  Standardize x with (μ, σ)
  for $m = 1$ to M **do**
    $a \leftarrow P_m(x)$; retrieve stored cell C_m with identifier a
    $\phi_m(x) \leftarrow$ the 36 descriptors of (x, C_m)
  end for
  $\Phi(x) \leftarrow [\phi_1(x), \ldots, \phi_M(x)]$
end for
return $\{\Phi(x)\}$

Table 2: Evaluation datasets after the accompanying loaders.

Dataset	ID rows	OOD rows	Features
Taxi	10,000	10,000	6
Electricity	9,986	10,014	6
Income	20,380	9,782	12
MVx6	20,384	20,384	9
California	10,315	10,319	7
ACS Accidents	22,653	3,955	45

rejection between an ID sample and an OOD sample. A mixture experiment with $5{,}000$ repetitions per contamination level records power as the OOD share grows.

The first point-wise experiment keeps the signed residual vector. Seven descriptors are computed: L_1, L_2, and L_∞ norms, counts above one, two, and three feature-wise standard deviations, and Mahalanobis distance. Logistic regression, a 50-tree random forest, and a depth-five decision tree are trained on these descriptors and, in a matched ablation, on the raw residual vector. Both variants use a stratified 70%–30% split with seed 42. Residual mean, covariance, and the descriptor scaler are estimated before this split in the residual-descriptor implementation. Those scores are therefore exploratory.

Partition protocol. ID and OOD rows are split independently by five-fold cross-validation with shuffling and seed 42. In each fold, partitions, cell statistics, and scalers are fitted on the ID training subset. Training labels of both classes are then balanced by random oversampling. Reported values are means over the five held-out folds. Reported standard deviations are the sample standard deviations of those five numbers. Pairwise Wilcoxon signed-rank tests are computed on the five fold-level ROC–AUC values. With five pairs the two-sided test cannot produce $p < 0.05$ when all signs agree, so the tests are used as a consistency check.

Representations and classifiers. Three inputs are compared: standardized raw attributes, the 36 meta-features, and their concatenation. Table 3 lists the fixed hyperparameters. Logistic regression uses balanced class weights and L-BFGS. The random forest uses 100 trees, log loss, and balanced class weights. The MLP uses hidden layers of 128 and 64 ReLU units, Adam, at most 500 iterations, learning rate 10^{-3}, and L_2 coefficient 10^{-4}.

Ablations. Each classifier is retrained on one meta-feature group at a time. A second ablation fits partitions on ID training rows only or on the union of labelled ID and OOD training rows. The ID-only variant is the operational method. Coefficient magnitudes of logistic regression and impurity importances of the random forest are averaged across partitions that share the same descriptor name.

Table 3: Fixed hyperparameters of the partition experiments.

Component	Setting
Quantile bins per feature	10
K-means clusters / K-means schemes	10 / $\max(1, \lfloor d/2 \rfloor)$
Randomized trees / maximum depth	5 / 3
Meta-features per partition	36
Cross-validation	5 folds, seed 42
Class balancing	random oversampling on the train fold
Logistic regression	L-BFGS, 500 iterations, balanced weights
Random forest	100 trees, log loss, balanced weights
MLP	$(128, 64)$ ReLU, Adam, $\mathrm{lr} = 10^{-3}$, $\alpha = 10^{-4}$

6 Experimental results

6.1 Autoencoder-based detection

The sample-wise AE–KS procedure yields empirical type-I error close to its intended level on all six datasets (Table 4). Four of the six 95% Wilson intervals contain $\alpha = 0.05$. The Electricity and California intervals lie slightly below 0.05. Comparisons of a pure ID sample with a pure OOD sample are rejected in every repetition, so the true-positive rate is 1.000 with Wilson interval $[0.9996, 1.0000]$. Power curves from the mixture experiment reach power of at least 0.8 when the evaluated sample contains approximately 15–20$\%$ OOD rows.

Table 4: Sample-wise AE–KS results from $10{,}000$ Monte Carlo repetitions. Each bootstrap sample contains $1{,}000$ observations and $\alpha = 0.05$.

Dataset	Empirical FPR	95% Wilson interval
Taxi	0.0462	[0.0423, 0.0505]
Electricity	0.0442	[0.0403, 0.0484]
Income	0.0474	[0.0434, 0.0517]
MVx6	0.0496	[0.0455, 0.0540]
California	0.0452	[0.0413, 0.0494]
ACS Accidents	0.0496	[0.0455, 0.0540]

Table 5 compares point-wise classifiers on residual descriptors with classifiers on the raw residual vector. The matched ablation was completed for Taxi, Electricity, and Income. The descriptor representation raises the best ROC–AUC on all three pairs, with increments of 0.152 on Taxi, 0.041 on Electricity, and 0.051 on Income. Across the descriptor experiment all six datasets reach ROC–AUC of at least 0.882. Training on noisy synthetic residual descriptors produced recall between 0.004 and 0.062 on six datasets. That run supports the later use of real labelled target rows.

Table 5: Point-wise residual-descriptor results. Each entry is the best test ROC–AUC among logistic regression, random forest, and decision tree. A dash marks an uncompleted raw-residual ablation.

Dataset	Raw residual vector	Residual descriptors
Taxi	0.7728	**0.9249**
Electricity	0.9584	**0.9993**
Income	0.8557	**0.9069**
MVx6	–	0.9784
California	–	0.8824
ACS Accidents	–	0.9986

Table 6: Five-fold mean ROC–AUC for logistic regression (LR) and random forest (RF) on standardized raw attributes and on the 36 meta-features. Bold marks the larger value in each LR or RF pair.

Dataset	LR raw	LR meta	RF raw	RF meta
Taxi	0.494	**0.998**	**1.000**	0.996
Electricity	0.803	**0.970**	**0.984**	0.977
Income	0.870	**0.917**	**0.914**	0.913
MVx6	**0.805**	0.785	**0.793**	0.788
California	**0.992**	0.990	**0.996**	0.994
ACS Accidents	0.523	**1.000**	1.000	1.000

Table 7: Logistic-regression ROC–AUC and the range of ROC–AUC across logistic regression, MLP, and random forest. Values are five-fold means.

Dataset	LR raw	LR meta	Range raw	Range meta
Taxi	0.494	0.998	0.506	0.002
Electricity	0.803	0.970	0.181	0.007
Income	0.870	0.917	0.044	0.014
MVx6	0.805	0.785	0.022	0.013
California	0.992	0.990	0.004	0.004
ACS Accidents	0.523	1.000	0.477	0.001

6.2 Linear classification on partition meta-features

Table 6 reports five-fold mean ROC–AUC for logistic regression and a random forest on raw attributes and on the 36 meta-features. Table 7 compresses the same runs into the logistic-regression score and the range of ROC–AUC across the three classifiers. Figure 2 is the intended visual companion of those two tables.

On Taxi, logistic regression on raw attributes has ROC–AUC 0.494, which is the level of a ranking that does not separate the classes. The same model on meta-features has ROC–AUC 0.998 ± 0.001. A random forest on raw attributes already has ROC–AUC 0.9999 on this pair, so the source and target tables are separable by axis-aligned splits of the original coordinates. The encoding reproduces that ranking with a linear rule. The ROC–AUC range across logistic regression, the MLP, and the random forest falls from 0.506 on raw attributes to 0.002 on meta-features.

ACS Accidents repeats the pattern. Logistic regression on raw attributes has ROC–AUC 0.523 ± 0.105. On meta-features the same model reaches 1.000. The random forest attains 1.000 on both inputs. The classifier range falls from 0.477 to 0.001.

Electricity and Income show the same movement for the linear model with a smaller margin. Logistic regression rises from 0.803 to 0.970 on Electricity and from 0.870 to 0.917 on Income. The random forest on raw attributes remains at 0.984 and 0.914 respectively. Concatenating raw attributes with meta-features changes logistic-regression ROC–AUC by at most 0.002 on these four pairs.

California is already solved by a linear rule on raw attributes (ROC–AUC 0.992). Meta-features preserve this level (0.990). MVx6 is the pair on which raw logistic regression (0.805) exceeds meta-feature logistic regression (0.785). The local cell statistics used here capture only part of the source–target change in that table. PR–AUC on MVx6 stays high (0.848 for logistic regression on meta-features), which indicates that ranking of the abundant OOD class remains possible while the ROC summary is moderate.

Fold-level Wilcoxon tests compare the five ROC–AUC values of meta-features with those of raw attributes. For logistic regression the five signs favour meta-features on Taxi, Electricity, Income, and ACS Accidents, and favour raw attributes on California and MVx6. The attained two-sided p-value is 0.0625 whenever all five signs agree. That value is the resolution limit of a five-fold signed-rank test.

Table 8 reports PR–AUC and F1 for the meta-feature representation. Taxi, Electricity, California, and ACS Accidents keep PR–AUC above 0.96 and F1 above 0.95 for at least one classifier. Income has PR–AUC 0.823 for logistic regression, consistent with an unequal ID/OOD ratio. MVx6 has

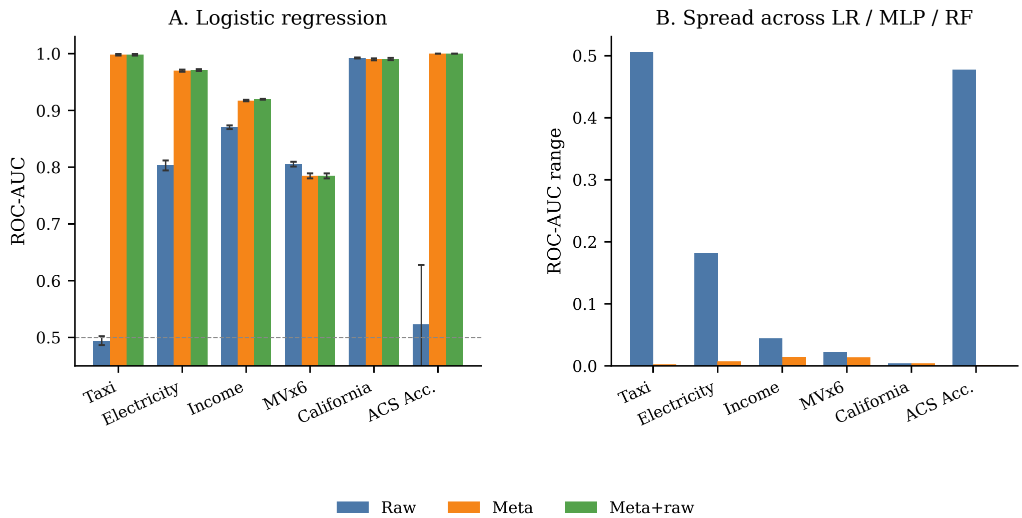

Figure 2: Linear classification quality and classifier range after the partition encoding. Panel A records logistic-regression ROC–AUC (five-fold mean $\pm$ standard deviation). Panel B records the spread among logistic regression, MLP, and random forest.

Table 8: Five-fold mean PR–AUC and F1 on the 36 meta-features without raw attributes. F1 uses the default probability threshold 0.5.

	LR		MLP		RF	
Dataset	PR	F1	PR	F1	PR	F1
Taxi	0.997	0.992	0.997	0.979	0.995	0.978
Electricity	0.962	0.915	0.963	0.919	0.972	0.920
Income	0.823	0.766	0.796	0.744	0.802	0.762
MVx6	0.848	0.686	0.834	0.688	0.850	0.708
California	0.989	0.950	0.990	0.951	0.993	0.958
ACS Accidents	1.000	1.000	0.999	0.999	1.000	1.000

PR–AUC 0.850 and F1 0.708 for the random forest. F1 uses the uncalibrated threshold 0.5, so it should be read together with PR–AUC.

6.3 Contribution of meta-feature groups

Table 9 reports the best five-fold mean ROC–AUC among the three classifiers when only one group is retained. The complete 36-feature run is included as a reference.

On Taxi the point-based group alone reaches 0.997, within 0.001 of the full encoding. Interaction-statistical descriptors follow at 0.921. The four remaining groups lie between 0.74 and 0.84. Electricity and California keep the same order: the point-based group stays within 0.004 of the full encoding. On ACS Accidents the point-based, interaction-statistical, and base groups all stay at or above 0.998, so several local statistics already separate that pair. All six restricted groups are reported for this dataset.

Income follows a different pattern. Marginal-statistical and interaction-informational groups reach 0.917 and 0.918, matching the full encoding, while the point-based group drops to 0.894. MVx6 likewise gives its highest single-group values to marginal and interaction descriptors (0.799 and 0.798), and the full concatenation (0.788) stays below those subsets. The source–target change in those two tables is aligned with shape and dependence of the cell, and the complete vector adds coordinates that the classifiers do not need.

Mean absolute logistic-regression coefficients, aggregated over partitions that share a descriptor name, agree with the Taxi and Electricity group scores. The largest magnitudes on Taxi belong to

Table 9: Best five-fold mean ROC–AUC among logistic regression, MLP, and random forest for each meta-feature group.

Group	Taxi	Elec.	Income	MVx6	Calif.	ACS
All 36	0.998	0.977	0.917	0.788	0.994	1.000
Point-based	0.997	0.974	0.894	0.787	0.993	1.000
Marginal statistical	0.767	0.960	0.917	0.798	0.969	0.973
Marginal informational	0.757	0.960	0.908	0.799	0.969	0.972
Interaction statistical	0.921	0.975	0.917	0.798	0.990	0.999
Interaction informational	0.756	0.960	0.918	0.798	0.969	0.972
Base	0.837	0.965	0.909	0.794	0.986	0.998

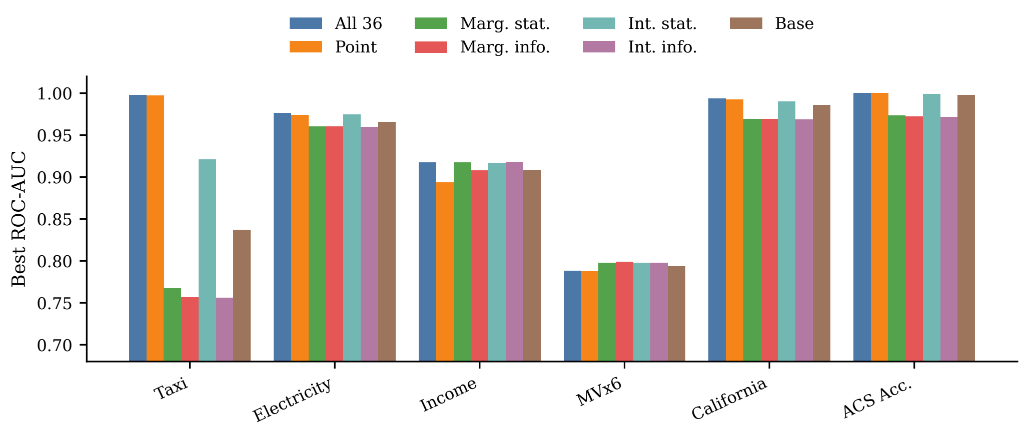

Figure 3: Best ROC–AUC by meta-feature group. The height of each bar is the maximum of the three classifier means on that restricted input.

`out_of_range_count`, diagonal Mahalanobis distance, and diagonal-Gaussian log-likelihood. The same three families dominate Electricity. On Income the coefficient mass is still concentrated on point-based names, while the group-level ROC–AUC shows that marginal and informational blocks already suffice for ranking. Coefficient magnitude and group ROC–AUC therefore answer related and distinct questions: the former describes the fitted linear rule, the latter describes a restricted feature set.

6.4 ID-only partition fitting

Table 10 compares the operational ID-only fit with an oracle that also sees OOD training rows when it builds cells. On Electricity, Income, MVx6, California, and ACS Accidents the two protocols stay within 0.01 ROC–AUC for logistic regression. On Taxi the ID-only fit has logistic-regression ROC–AUC 0.998 and the oracle has 0.990. The operational protocol is therefore sufficient for the pairs studied here. The earlier holdout experiment that used ten meta-features and a single split had shown a large Taxi gain for the oracle. That gain does not appear under five-fold evaluation of the full taxonomy.

7 Conclusion and discussion

The study asked three questions, and the experiments give the following answers.

On question 1, the AE–KS batch test keeps empirical type-I error near $\alpha = 0.05$ on six pairs and rejects every pure-OOD sample of size $1{,}000$. Mixtures with an OOD share of about 15–20$\%$ reach power 0.8 in the mixture experiment. The test flags a changed batch. It leaves the contributing rows unidentified.

Table 10: Five-fold mean ROC–AUC of logistic regression when partitions are fitted on ID training rows only or on the union of ID and OOD training rows.

Dataset	Fit on ID	Fit on ID and OOD
Taxi	0.998	0.990
Electricity	0.970	0.944
Income	0.917	0.926
MVx6	0.785	0.794
California	0.990	0.992
ACS Accidents	1.000	1.000

On question 2, norms, exceedance counts, and Mahalanobis distance of the residual vector raise the best ROC–AUC on the three pairs with a matched raw-residual ablation. The comparison is exploratory because residual reference statistics were computed before the classification split.

On question 3, the 36 meta-features give logistic regression ROC–AUC 0.998 on Taxi and 1.000 on ACS Accidents and shrink the classifier range to 0.002 and 0.001 on those pairs. Income shows a smaller linear gain, from 0.870 to 0.917. Group scores and coefficient magnitudes assign Taxi, Electricity, California, and ACS Accidents to point-to-cell geometry, and assign Income and MVx6 to marginal and dependence statistics. MVx6 is the pair on which raw logistic regression keeps a higher ROC–AUC than the partition encoding. The encoding is therefore a linear, named coordinate system for an observed source–target shift, with a group-level account of which local statistics move.

Limitations. Training uses labelled target rows from the same pair that appears at test time. The numbers measure discrimination of that pair. Transfer to an OOD type withheld from training is untested. Taxi and ACS Accidents are already separated by a random forest on raw attributes, so their perfect scores describe linear recoverability of an easy ranking. The partition-family sweep is absent from the five-fold tables. Computational cost grows with d because the number of partitions is $M = d + \max(1, \lfloor d/2 \rfloor) + 5$ and several descriptors use pairwise information. External detectors such as ECOD, COPOD, Isolation Forest, DevNet, or Deep SAD are not run on these splits. The internal baselines isolate the contribution of the representation under a shared classifier family.

Future work. Leave-one-dataset-out evaluation is a natural next measurement, because the 36 descriptors have the same meaning on every table once they are aggregated over partitions. A labelled-OOD budget curve would place the encoding next to weakly supervised detectors. Nested selection of the number of bins, clusters, and trees, together with sparse descriptor selection, would reduce cost on wide tables.

8 Code and data availability

The implementation, dataset loaders, and experiment runner live at `https://github.com/BodBodBod/ood_detection_itmo`. The runner writes fold-level and summary CSV files to `results/`. The six pairs reported in Section 5 were produced by `point_wise_partition_meta.run_experiments` with tasks `representation`, `id_ood`, `importance`, and `groups`. Dependencies are listed in `requirements.txt`.

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A Meta-feature taxonomy

The representation uses the following 36 descriptors, computed on the standardized ID training cell C of a query x.

Point-based. `dist_to_mean`: Euclidean distance from x to the cell mean. `normalized_dist`: that distance divided by a cell radius. `cosine_dist`: cosine distance to the cell mean. `dist_to_median`: Euclidean distance to the cell median. `log_likelihood`: sum of diagonal-Gaussian log-densities. `local_outlier_factor_score`: mean kNN distance of x divided by the mean kNN distance of ID rows in C. `knn_distance_k`: distance to the k-th neighbour in C. `out_of_range_count`: number of coordinates of x outside the coordinate-wise min–max range of C.

Marginal statistical. Norms of the cell mean, standard deviation, skewness, kurtosis, median absolute deviation, interquartile range, and 10%-trimmed mean.

Marginal informational. Joint histogram entropy of C, mean and standard deviation of per-feature entropies, mean KL divergence of cell margins to the ID training margins, and mean quantile surprisal of x.

Interaction statistical. Diagonal Mahalanobis distance, covariance trace, log-determinant, condition number, mean absolute Pearson correlation, and mean absolute Spearman correlation.

Interaction informational. Mean and maximum pairwise mutual information, Gaussian total correlation, mean pairwise joint entropy, and mean pairwise Jensen–Shannon divergence.

Base. $\log(1 + |C|)$, $|C|/n$, the number of coordinates of x beyond two cell standard deviations, the number of partitions that place x in a cell whose density is at least the median density, and tree-leaf depth (zero on non-tree partitions).